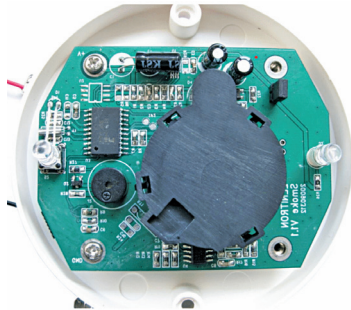


Interesting

Reference Designs OF SMOKE DETECTORS

Title of Reference Design	Wireless Smoke Detector	Smoke Detector
Image		
Key Attractions	The wireless smoke detector design offers network-wide alarm triggering, easy setup, long battery life, and remote access, with a sub-GHz wireless module for communication and powered by a single battery for enhanced safety and convenience.	The smoke detector design features a photoelectric sensor, high-decibel alarm, and battery operation with low power consumption. It provides reliable fire detection, ensuring safety with a wide coverage area and advanced power management.
Highlights	The wireless smoke detector reference design from Microchip is centred around a low-power nanoWatt XLP 8-bit PIC microcontroller, the PIC16LF1825. The RE46C190 photoelectric smoke detector IC is utilised to provide all necessary features for a photoelectric smoke detector with minimal external components. Alarm conditions are communicated through the MRF89XAM8A (or 9A) sub-GHz wireless module. The board is powered by a single CR123 battery. The key features of the system include an alarm event that triggers all alarms in the network for enhanced safety. The setup is easy with automatic wireless node detection, and the system is powered by a single CR123 3V battery. It features low power sleep modes, including a 50nA sleep mode for the PIC microcontroller, 0.5µA LPWDT operation in sleep mode, and a 0.1µA sleep mode for the transceiver module, leading to a long battery life. Remote access capability allows users to receive remote alarm alerts, test alarm functionality, and monitor battery voltage from a distance.	The Cypress reference design serves as a smoke detector that utilises a photoelectric sensor to detect infrared radiation refracted by smoke or dust particles produced by fire. It features a high-decibel alarm exceeding 70dB within a 3m range, ensuring adequate coverage of a 60 square metre area when mounted less than 5m above the floor. The detector is powered by two batteries operating at a 1.8V to 3.3V DC voltage range and includes a low battery output warning. The DC-DC boost converter maintains a constant power output of 3.3V when the battery voltage falls below 3V, using the EMH7600 boost converter that supports an input range of 0.3V to 6V and provides a 3.3V output with up to 500mA. The design also includes a photoelectric detector with an IR-emitting diode and an IR-receiving diode positioned within a dark cavity, requiring an amplifier circuit to amplify the signal for the analogue-to-digital stage. To reduce power consumption, the design implements a sleep mode with periodic wakeups, which is crucial in battery-powered systems from both a firmware and hardware perspective.
Applications	The reference design is suitable for early fire detection and home safety.	The reference design is suitable for residential homes, hotels, etc.
OEM Brand	Microchip	Cypress
URL	https://www.electronicsforu.com/electronics-projects/wireless-smoke-detector-reference-design	https://www.electronicsforu.com/electronics-projects/smoke-detector-reference-design